



Symmetry/Topological Order Correspondence

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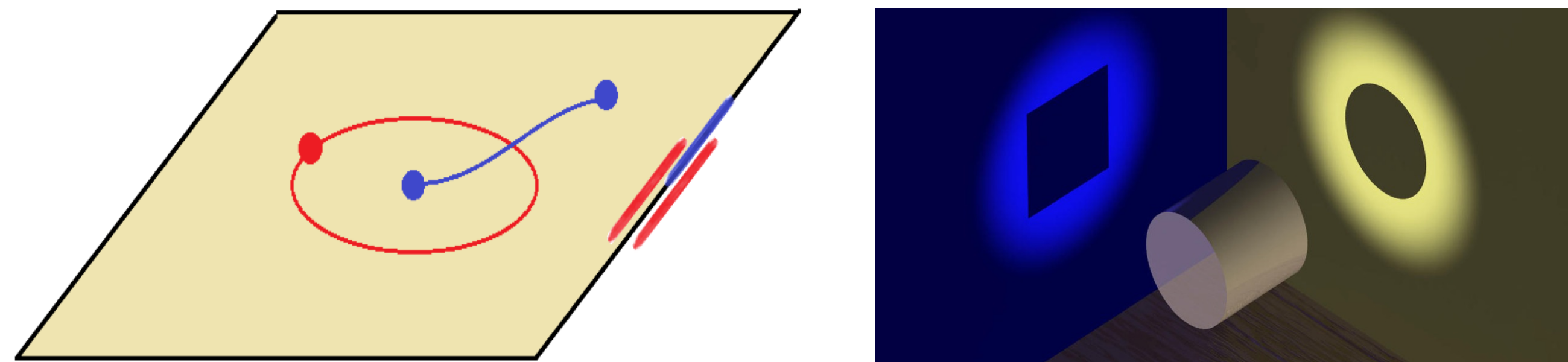


Motivation

Menagerie of symmetries – **anomalous**, **higher-form**, **higher group**, **algebraic/non-invertible**

Question: How to organize and **unify** our understanding of all these symmetries?

Proposal: Topological Order (TO) in one higher dimension – we dub this *categorical symmetry*^[1]



Algebra of Patch Operators

Local Symmetric Operators describe symmetry without reference to symmetry transformations

Example: $\mathbb{Z}_2$ symmetry in 1+1D TFIM

LSOs generated by $\{X_i, Z_i Z_{i+1}\}$ commute with $W = \prod_i X_i$

Algebra of LSOs $\supset$ **Transparent Patch Operators**

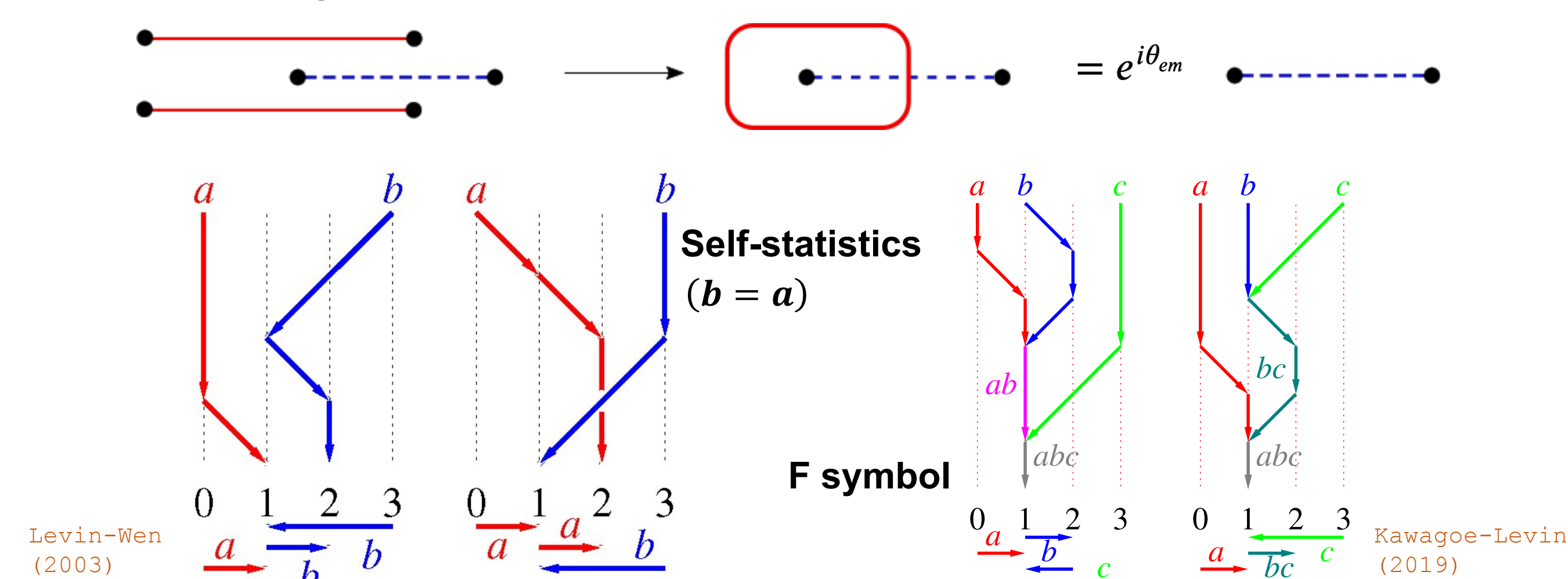
$$\begin{array}{c} \bullet \text{---} \bullet \\ \bullet \text{---} \bullet \end{array} = \begin{array}{c} \bullet \text{---} \bullet \\ \bullet \text{---} \bullet \end{array} \quad \begin{array}{c} i \quad j \\ \bullet \text{---} \bullet \\ j \quad k \end{array} = \begin{array}{c} i \quad k \\ \bullet \text{---} \bullet \end{array}$$

"invisible" bulk **joining condition**

Endpoints interpreted as particles

$$Z_{strij} = Z_i Z_j \sim e_i e_j \quad X_{strij} = X_i X_{i+1} \cdots X_{j-1} \sim m_i m_j$$

Operator algebra encodes data of 2+1D TO^[2]



Dualities and Phase Transitions

Isomorphic Patch Operator Algebra $\Rightarrow$ Duality

Example: Mixed anomalous $\mathbb{Z}_2 \times \mathbb{Z}_2 \cong$ anomaly-free $\mathbb{Z}_4$ (1+1D)

Patch Operators

$$Z_{strij} = Z_i Z_j, \quad W_{strij} = \begin{array}{c} i \quad j \\ \text{---} \\ \tilde{O} \quad X \quad X \quad X \quad \tilde{O}^\dagger \\ X \quad X \quad X \quad X \end{array}$$

$$\tilde{Z}_{strij} = \tilde{Z}_i \tilde{Z}_j, \quad \tilde{W}_{strij} = \begin{array}{c} \tilde{X} \quad \tilde{X} \quad \tilde{X} \quad \tilde{X} \\ \text{---} \\ \tilde{s} \quad \tilde{s} \quad \tilde{s} \quad \tilde{s} \end{array}$$

$$s_{i,i+1} = \exp\left(\frac{i\pi}{4}(Z_{i+1} - Z_i)(\tilde{Z}_i + 1)\right) \quad \tilde{O}_j = \frac{1}{\sqrt{2}}(1 - i\tilde{Z}_j)$$

Operator Correspondence [*Categorical Symmetry* = $\mathcal{D}(\mathbb{Z}_4)$]

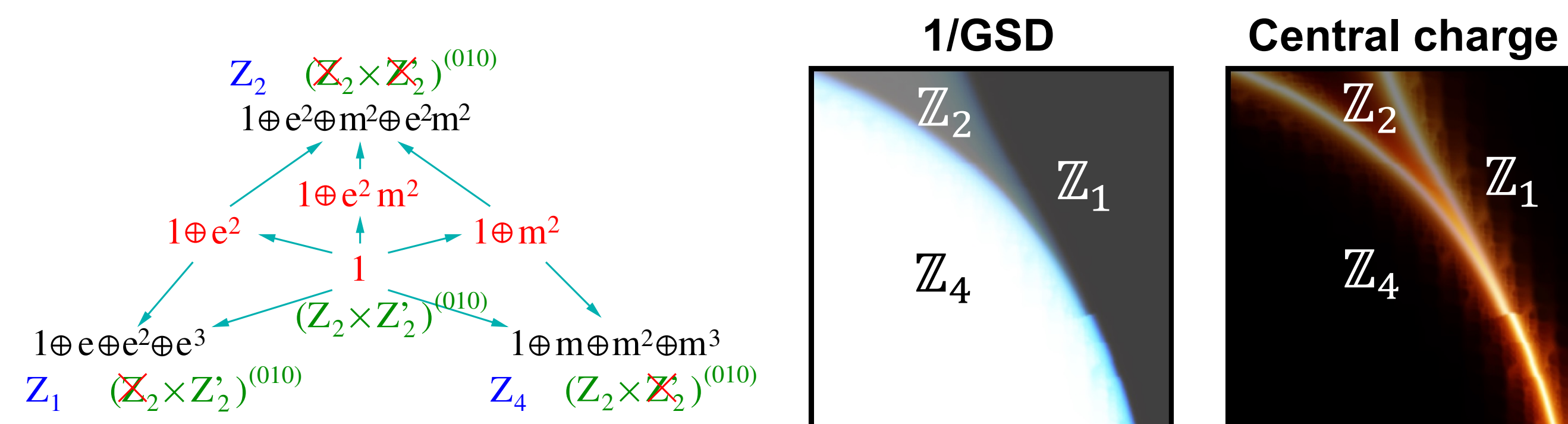
$\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry with mixed anomaly	Anomaly-free $\mathbb{Z}_4$ symmetry
$W_{strij} = \tilde{O}_i X_i X_{i+1} \cdots X_{j-1} \tilde{O}_j^\dagger$	$\tilde{O}_i \tilde{Z}_i \tilde{O}_j^\dagger \tilde{Z}_j$
$\tilde{W}_{strij} = \tilde{X}_{i+1} \cdots \tilde{X}_j s_{i,i+1} \cdots s_{j-1,j}$	$\prod_{k=i}^j X_k C X(\tilde{g}_k, \tilde{g}_k) = \prod_{k=i}^j U_k^{-1}$
$\tilde{Z}_{strij} = \tilde{Z}_i \tilde{Z}_j$	$\tilde{Z}_i \tilde{Z}_j$
$Z_{strij} = Z_i Z_j$	$\prod_{k=i+1}^j \tilde{X}_k = \prod_{k=i+1}^j U_k^2$

Condensable Algebras

Gapped phases of system $\mathcal{C} \leftrightarrow$ gapped boundaries of $\mathcal{M}(\mathcal{C})$
 $\leftrightarrow$ Lagrangian condensable algebras

Gapless states of $\mathcal{C} \leftrightarrow$ non-Lagrangian condensable algebras

Example: $\mathbb{Z}_4 \cong (\mathbb{Z}_2 \times \mathbb{Z}_2)^{\omega_{III}}$ phase diagram

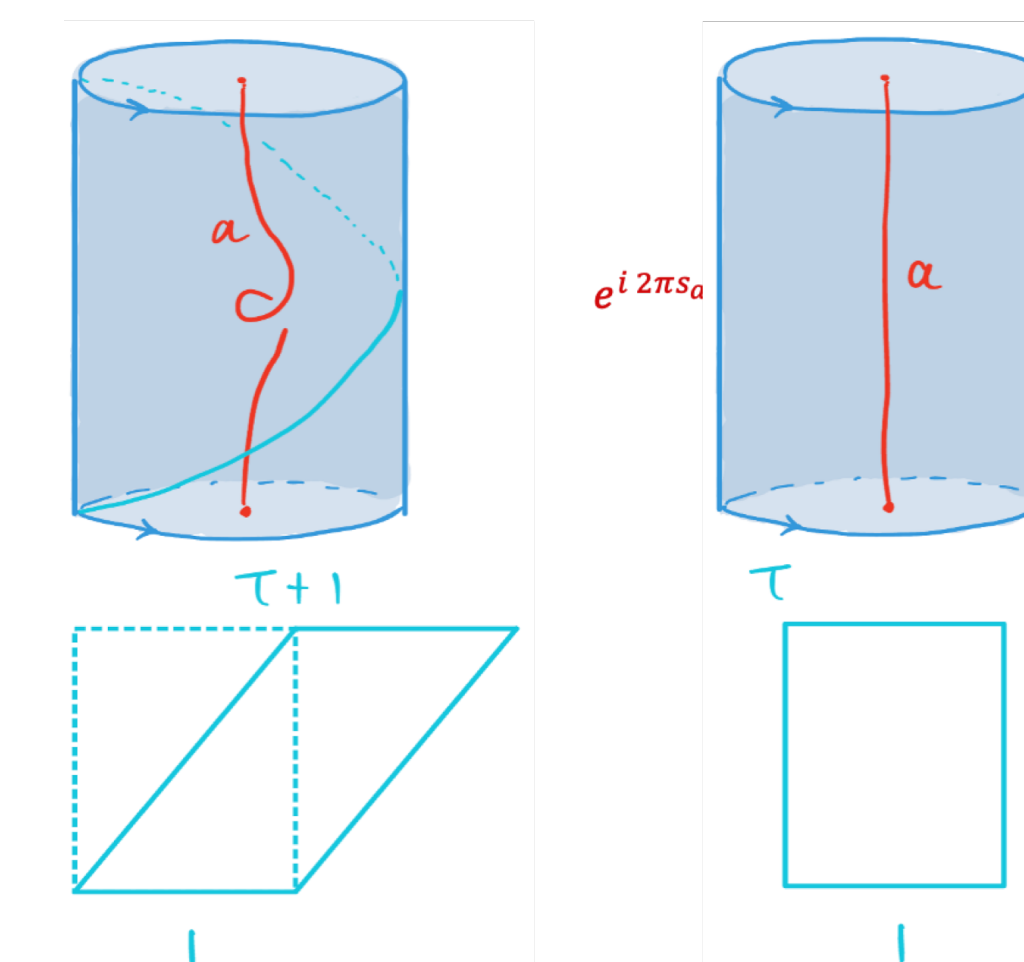


Multi-component Bootstrap^[3]

$$T_{ab}^{\mathcal{M}} Z_b(\tau) = Z_a(\tau + 1)$$

$$S_{ab}^{\mathcal{M}} Z_b(\tau) = Z_a(-1/\tau)$$

1—condensed gapless boundaries have $Z_{a \neq 1} \neq 0$



Maximal Categorical Symmetry

Gapless state described by modular invariant 1+1D CFT – Partition fn. decomposed into anyon sectors of the bulk TO^[3]

$$\begin{array}{c} \tilde{\mathcal{R}} \\ \text{---} \\ \mathcal{M} \\ \text{---} \\ CFT_{ano} \end{array} \left. \vphantom{\begin{array}{c} \tilde{\mathcal{R}} \\ \text{---} \\ \mathcal{M} \\ \text{---} \\ CFT_{ano} \end{array}} \right\} CFT_{af} = CFT_{ano} \boxtimes_{\mathcal{M}} \tilde{\mathcal{R}}$$

$$\begin{array}{c} \tilde{\mathcal{R}}' \quad \tilde{\mathcal{R}} \\ \text{---} \\ \mathcal{M}' \quad \mathcal{M} \\ \text{---} \\ CFT_{ano}' \quad CFT_{ano} \quad CFT_{inv} \end{array} \beta$$

Many such decompositions may be possible

The decomposition in terms of the *largest* TO of them all defines the **maximal categorical symmetry**^[4]

Example:

Ising model: $\mathcal{D}(\mathbb{Z}_2)$ *categorical symmetry*

Kramers-Wannier duality symmetry emerges at critical point

Maximal *categorical symmetry* = $\mathcal{D}(\text{Ising})$

$$\begin{array}{c} \text{vec}_{\mathbb{Z}_2} \\ \text{---} \\ \mathcal{D}(\mathbb{Z}_2) \\ \text{---} \\ Is_{\mathcal{D}(\mathbb{Z}_2)} \end{array} \left. \vphantom{\begin{array}{c} \text{vec}_{\mathbb{Z}_2} \\ \text{---} \\ \mathcal{D}(\mathbb{Z}_2) \\ \text{---} \\ Is_{\mathcal{D}(\mathbb{Z}_2)} \end{array}} \right\} Is_{inv} \left\{ \begin{array}{c} \mathcal{F}_{Is} \\ \text{---} \\ \mathcal{D}(Is) \\ \text{---} \\ Is_{\mathcal{D}(Is)} \end{array} \right. Is_{af}$$

Proposal: Maximal *categorical symmetry* captures all **emergent** symmetries and largely characterizes the gapless state

cf. Chang-Lin-Shao-Wang-Yin (2018)

Future Directions

1. Extend symmetry/TO correspondence to higher dimensions: e.g. 3+1D bosonic TOs are *just* DW theories
2. Maximal *categorical symmetry* systematics: symmetry twists as a tool to reveal emergent symmetries/anomalies
3. *Categorical symmetry* beyond TFT
4. Understanding of other dualities in terms of Symmetry/TO correspondence

1. Ji, W. and Wen, X.G., 2020. Categorical symmetry and noninvertible anomaly in symmetry-breaking and topological phase transitions. *Physical Review Research*, 2(3), p.033417.
2. Chatterjee, A. and Wen, X.G., 2023. Symmetry as a shadow of topological order and a derivation of topological holographic principle. *Physical Review B*, 107(15), p.155136.
3. Ji, W. and Wen, X.G., 2019. Noninvertible anomalies and mapping-class-group transformation of anomalous partition functions. *Physical Review Research*, 1(3), p.033054.
4. Chatterjee, A., Ji, W. and Wen, X.G., 2022. Emergent maximal categorical symmetry in a gapless state. arXiv preprint arXiv:2212.14432.